

|                   |          |          |          |
|-------------------|----------|----------|----------|
|                   | $c_1$    | $c_2$    | $c_3$    |
| $r_1 \rightarrow$ | $x_{11}$ | $x_{12}$ | $x_{13}$ |
| $r_2 \rightarrow$ | $x_{21}$ | $x_{22}$ | $x_{23}$ |
| $r_3 \rightarrow$ | $x_{31}$ | $x_{32}$ | $x_{33}$ |

Alice: Assigned  $j$ , and must fill in  $r_j$

Bob: Assigned  $k$ , and must fill in  $c_k$

$x_{jk} \in \{1, -1\}$  To win: Alice must fill  $a_{jk}$  so that:  $\prod_k a_{jk} = 1$  — Condition 1

Bob must fill  $b_{jk}$  so that:  $\prod_j b_{jk} = 1$  — Condition 2

And  $a_{jk} = b_{jk}$  — Condition 3

$$\left\{ \begin{array}{l} \begin{array}{ccc|l} x_{11} & x_{12} & x_{13} & = 1 \\ x_{21} & x_{22} & x_{23} & = 1 \\ x_{31} & x_{32} & x_{33} & = 1 \\ \hline -1 & -1 & -1 & \end{array} \\ \begin{array}{l} x_{11}x_{12}x_{13} = 1 \\ x_{21}x_{22}x_{23} = 1 \\ x_{31}x_{32}x_{33} = 1 \end{array} \end{array} \right. \quad \left\{ \begin{array}{l} x_{11}x_{21}x_{31} = -1 \\ x_{12}x_{22}x_{32} = -1 \\ x_{13}x_{23}x_{33} = -1 \end{array} \right.$$

$$\prod_{\text{all}} x = 1^3 \quad \prod_{\text{all}} x = (-1)^3 = -1$$

False, can't satisfy constraints with numbers as multiplication of numbers is commutative

Best classical strategy

|    |    |         |           |
|----|----|---------|-----------|
| 1  | 1  | 1       | $\cdot 1$ |
| 1  | 1  | 1       | $\cdot 1$ |
| -1 | -1 | $\pm 1$ | $\cdot 1$ |
| -1 | -1 | -1      |           |

1. Select a  $2 \times 2$  spot to fill in a predetermined values)  
(ex. upper left  $2 \times 2$ )

2. Complete the missing spots in accordance with the constraints

Condition 1, 2: Full filled

Condition 3: Uncertain

3. There will be 1 square which will violate the third rule  
but that will only be a concern when  $j=3, k=3$

$\therefore$  Prob of winning =  $8/9$

Quantum strategy

Operator theory

Commutator:  $[a, b] = ab - ba$

Anti Commutator:  $\{a, b\} = ab + ba$

|          |          |          |       |
|----------|----------|----------|-------|
| $B_{11}$ | $B_{12}$ | $B_{13}$ | $= I$ |
| $B_{21}$ | $B_{22}$ | $B_{23}$ | $= I$ |
| $B_{31}$ | $B_{32}$ | $B_{33}$ | $= I$ |
| $-I$     | $-I$     | $-I$     |       |

Constraints:  $\prod_k B_{jk} = I$

$\prod_j B_{jk} = -I$

$$\left\{ \begin{array}{l} [B_{jk_1}, B_{jk_2}] = 0 \quad \text{same row} \\ [B_{j_1k}, B_{j_2k}] = 0 \quad \text{same col} \end{array} \right.$$

For compatibility of measurements: the operators in a row/col must commute ex.  $[B_{12}, B_{13}] = 0$   
 $B_{12}B_{13} = B_{13}B_{12}$

Possible to construct this table with Pauli matrices:

$$B_{jk} \longleftrightarrow \sigma_j \otimes \sigma_k$$

Pauli matrices:  $\sigma_I \cdot I$ ,  $\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ ,  $\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ,  $\sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$

$$\{\sigma_a, \sigma_b\} = 2\delta_{ab}, \quad [\sigma_a \otimes \sigma_b, \sigma_b \otimes \sigma_a] = 0$$

i.e.

$$1. \sigma_a \sigma_b = -\sigma_b \sigma_a$$

and

$$2. \sigma_a \sigma_a = I = \sigma_a^2$$

ex.

$$[\sigma_z \otimes \sigma_y, \sigma_y \otimes \sigma_z] = (\sigma_z \otimes \sigma_y)(\sigma_y \otimes \sigma_z) - (\sigma_y \otimes \sigma_z)(\sigma_z \otimes \sigma_y)$$

From Kronecker-mixed-product property:

$$(A \otimes B)(C \otimes D) = AC \otimes BD$$

$$\therefore = (\sigma_z \sigma_y \otimes \sigma_y \sigma_z) - (\sigma_y \sigma_z \otimes \sigma_z \sigma_y)$$

$$\therefore \sigma_a \sigma_b = -\sigma_b \sigma_a$$

$$\therefore = (\sigma_z \sigma_y \otimes \sigma_y \sigma_z) - (+\sigma_z \sigma_y \otimes +\sigma_y \sigma_z)$$

$$= (\sigma_z \sigma_y \otimes \sigma_y \sigma_z) - (\sigma_z \sigma_y \otimes \sigma_y \sigma_z)$$

$$= 0$$

$$\begin{aligned} \sigma_x \sigma_y &= -\sigma_y \sigma_x = i\sigma_z \\ \sigma_y \sigma_z &= -\sigma_z \sigma_y = i\sigma_x \\ \sigma_z \sigma_x &= -\sigma_x \sigma_z = i\sigma_y \end{aligned}$$

↓

$$\begin{aligned} \therefore \sigma_x \sigma_y \sigma_y \sigma_z \sigma_z \sigma_x &= -i\sigma_z \sigma_x \sigma_y \\ \cancel{\sigma_x \sigma_y} \cancel{\sigma_z \sigma_x} &= i(\cancel{\sigma_x \sigma_z}) \sigma_y \\ \cancel{\sigma_x^2} &= -i\sigma_x \sigma_y \sigma_z \end{aligned}$$

$$\therefore -i\sigma_x \sigma_y \sigma_z = I = \sigma_a^2$$

Construct table using Paulis

$$\begin{array}{|c|c|c|} \hline \sigma_I \otimes \sigma_z & \sigma_x \otimes \sigma_I & \sigma_x \otimes \sigma_z \\ \hline \sigma_z \otimes \sigma_I & \sigma_I \otimes \sigma_x & \sigma_z \otimes \sigma_x \\ \hline -\sigma_z \otimes \sigma_z & -\sigma_x \otimes \sigma_x & -\sigma_y \otimes \sigma_y \\ \hline \end{array} \begin{array}{l} = I \\ = I \\ = I \end{array}$$

1                  2                  3

$-I \quad -I \quad -I$

1. Select a 2x2 spot to fill in a predetermined values)  
(ex. upper left 2x2)

2. Complete the missing spots in accordance with the constraints

\* From mixed-product  
(A ⊗ B)(C ⊗ D) = AC ⊗ BD

↓

$$r_1: (\sigma_I \otimes \sigma_z)(\sigma_x \otimes \sigma_I)(\sigma_x \otimes \sigma_z) = I\sigma_x \sigma_x \otimes \sigma_z I\sigma_z = \sigma_x^2 \otimes \sigma_z^2 = I \checkmark$$

$$r_2: (\sigma_z \otimes \sigma_I)(\sigma_I \otimes \sigma_x)(\sigma_z \otimes \sigma_x) = \sigma_z I\sigma_z \otimes I\sigma_x \sigma_x = \sigma_z^2 \otimes \sigma_x^2 = I \checkmark$$

$$c_1: (\sigma_I \otimes \sigma_z)(\sigma_z \otimes \sigma_I)(-\sigma_z \otimes \sigma_z) = -I\sigma_z \sigma_z \otimes \sigma_z I\sigma_z = -\sigma_z^2 \otimes \sigma_z^2 = -I \checkmark$$

$$c_2: (\sigma_x \otimes \sigma_I)(\sigma_I \otimes \sigma_x)(-\sigma_x \otimes \sigma_x) = -\sigma_x I\sigma_x \otimes I\sigma_x \sigma_x = -\sigma_x^2 \otimes \sigma_x^2 = -I \checkmark$$

$$r_3: -(\sigma_z \otimes \sigma_z)(\sigma_x \otimes \sigma_x)(\sigma_y \otimes \sigma_y) = -\sigma_z \sigma_x \sigma_y \otimes \sigma_z \sigma_x \sigma_y = -\sigma_x \sigma_y \sigma_z \otimes \sigma_x \sigma_y \sigma_z = -iI \otimes iI = I \checkmark$$

From  $\sigma_a \sigma_b = -\sigma_b \sigma_a$

$$c_3: (\sigma_x \otimes \sigma_z)(\sigma_z \otimes \sigma_x)(-\sigma_y \otimes \sigma_y) = -\sigma_x \sigma_z \sigma_y \otimes \sigma_z \sigma_x \sigma_y = \sigma_x \sigma_y \sigma_z \otimes \sigma_x \sigma_y \sigma_z = iI \otimes iI = -I \checkmark$$

From  $\sigma_a \sigma_b = -\sigma_b \sigma_a$       From  $\sigma_x \sigma_y \sigma_z = iI$

↳ Condition 1, 2: Fulfilled

On intersection, to win, both must agree

$$\therefore H_{\text{measure}} |\psi_{q_{0,1,2,3}}\rangle = \lambda |\psi_{q_{0,1,2,3}}\rangle$$

measure same  $\therefore \lambda_{\text{Alice}} = \lambda_{\text{Bob}}$

$$\therefore \lambda = \{1, -1\} \quad \text{for } \lambda_{\text{Alice}} = \lambda_{\text{Bob}} \therefore \lambda_{\text{Alice}} \lambda_{\text{Bob}} = 1 \quad \begin{matrix} \nearrow 1, 1 \\ \searrow -1, -1 \end{matrix}$$

$$\therefore H_{\text{measure}} |\psi_{q_{0,1,2,3}}\rangle = (1) |\psi_{q_{0,1,2,3}}\rangle$$

$$(B_{jk})_{q_{0,1}} \otimes (B_{jk})_{q_{1,2}} |\psi_{q_{0,1,2,3}}\rangle ; B_{jk} = (\sigma_a \otimes \sigma_b)$$

$$(\sigma_a \otimes \sigma_b)_{q_{0,1}} \otimes (\sigma_a \otimes \sigma_b)_{q_{1,2}} |\psi_{q_{0,1,2,3}}\rangle$$

$$(\sigma_a \otimes \sigma_b \otimes \sigma_a \otimes \sigma_b) |\psi\rangle$$

When  $j, k \in \{1, 2\}$

|                   |                              |                              |                              |
|-------------------|------------------------------|------------------------------|------------------------------|
|                   | $c_1$<br>$\downarrow$        | $c_2$<br>$\downarrow$        |                              |
| $r_1 \rightarrow$ | $\sigma_I \otimes \sigma_z$  | $\sigma_x \otimes \sigma_I$  | $\sigma_x \otimes \sigma_z$  |
| $r_2 \rightarrow$ | $\sigma_z \otimes \sigma_I$  | $\sigma_I \otimes \sigma_x$  | $\sigma_z \otimes \sigma_x$  |
|                   | $-\sigma_z \otimes \sigma_z$ | $-\sigma_x \otimes \sigma_x$ | $-\sigma_y \otimes \sigma_y$ |

Prepare 2 pairs of  $|\Phi^+\rangle$  bell states  $\rightarrow$

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

$$|\psi\rangle = |\Phi^+\rangle \otimes |\Phi^+\rangle = |1_3 1_2 1_1 0_0\rangle$$

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \otimes \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$|\psi\rangle = \frac{1}{2} (|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle)$$

$$\therefore |q_3 q_2 q_1 q_0\rangle$$

Alice possess  $1_3, 1_1$ , Bob possess  $1_2, 1_0$

|                              |                              |                              |                              |                             |
|------------------------------|------------------------------|------------------------------|------------------------------|-----------------------------|
| $q_3$                        | $q_1$                        | $q_3$                        | $q_1$                        | $\sigma_x \otimes \sigma_z$ |
| $\sigma_I \otimes \sigma_z$  | $\sigma_x \otimes \sigma_I$  | $\sigma_x \otimes \sigma_I$  | $\sigma_I \otimes \sigma_x$  |                             |
| $q_2$                        | $q_0$                        | $q_2$                        | $q_0$                        | $\sigma_z \otimes \sigma_x$ |
| $\sigma_z \otimes \sigma_I$  | $\sigma_I \otimes \sigma_x$  | $\sigma_I \otimes \sigma_x$  | $\sigma_z \otimes \sigma_x$  |                             |
| $q_2$                        | $q_0$                        | $q_2$                        | $q_0$                        |                             |
| $-\sigma_z \otimes \sigma_z$ | $-\sigma_x \otimes \sigma_x$ | $-\sigma_x \otimes \sigma_x$ | $-\sigma_y \otimes \sigma_y$ |                             |



If measured in the same basis  $(x, x), (y, y), (z, z)$

$$q_3 = q_2, q_1 = q_0$$

$r_1$ : Alice measure  $q_3$  in  $\sigma_I \sigma_x = \underline{\sigma_x}$  basis  $\rightarrow$  result for  $x_{12}$   
measure  $q_1$  in  $\sigma_z \sigma_I = \underline{\sigma_z}$  basis  $\rightarrow$  result for  $x_{11}$

$c_1$ : Bob measure  $q_2$  in  $\sigma_I \sigma_z = \underline{\sigma_z}$  basis  $\rightarrow$  result for  $x_{21}$   
measure  $q_0$  in  $\sigma_z \sigma_I = \underline{\sigma_z}$  basis  $\rightarrow$  result for  $x_{11}$

$r_2$ : Alice's measure  $q_3$  in  $\sigma_z \sigma_z = \sigma_z$  basis  $\rightarrow$  result for  $x_{21}$   
 measure  $q_1$  in  $\sigma_z \sigma_x = \sigma_x$  basis  $\rightarrow$  result for  $x_{22}$

$C_2$ : Bob's measure  $q_2$  in  $\sigma_x \sigma_z = \sigma_x$  basis  $\rightarrow$  result for  $x_{12}$   
 measure  $q_0$  in  $\sigma_z \sigma_x = \sigma_x$  basis  $\rightarrow$  result for  $x_{22}$

Scenario 1, Row 1, Col 1:  $|\psi\rangle = \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle)$   
 $(0, 0)$

$$U|\psi\rangle = (H \otimes I \otimes I \otimes I)|\psi\rangle$$

$$= \frac{1}{2} \left[ (H \otimes I \otimes I \otimes I)(|0\rangle \otimes |0\rangle \otimes |0\rangle \otimes |0\rangle) \right. \\
+ (H \otimes I \otimes I \otimes I)(|0\rangle \otimes |0\rangle \otimes |1\rangle \otimes |1\rangle) \\
+ (H \otimes I \otimes I \otimes I)(|1\rangle \otimes |1\rangle \otimes |0\rangle \otimes |0\rangle) \\
\left. + (H \otimes I \otimes I \otimes I)(|1\rangle \otimes |1\rangle \otimes |1\rangle \otimes |1\rangle) \right]$$

$$= \frac{1}{2} (|+000\rangle + |+011\rangle + |-100\rangle + |-111\rangle)$$

$$= \frac{1}{2\sqrt{2}} (|0000\rangle + |1000\rangle + |0011\rangle + |1011\rangle \\
+ |0100\rangle - |1100\rangle + |0111\rangle - |1111\rangle)$$

|                |       |  |
|----------------|-------|--|
| $q_1$<br>$q_4$ | $q_3$ |  |
| $q_2$          |       |  |
|                |       |  |

All possible states  
 have  $q_1 = q_4$

$\therefore$  Constant

Scenario 2, Row 1, Col 2:  $|\psi\rangle = \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle)$   
 $(0, 1)$

$$U|\psi\rangle = (H \otimes H \otimes I \otimes H)|\psi\rangle$$

$$= \frac{1}{2} \left[ (H \otimes H \otimes I \otimes H)(|0\rangle \otimes |0\rangle \otimes |0\rangle \otimes |0\rangle) \right. \\
+ (H \otimes H \otimes I \otimes H)(|0\rangle \otimes |0\rangle \otimes |1\rangle \otimes |1\rangle) \\
+ (H \otimes H \otimes I \otimes H)(|1\rangle \otimes |1\rangle \otimes |0\rangle \otimes |0\rangle) \\
\left. + (H \otimes H \otimes I \otimes H)(|1\rangle \otimes |1\rangle \otimes |1\rangle \otimes |1\rangle) \right]$$

$$= \frac{1}{2} (|++0\rangle + |+1-\rangle + |-0+\rangle + |--1-\rangle)$$

$$\downarrow$$

$$= \frac{1}{4\sqrt{2}} (0000 + 0001 + 0100 + 0101 + 1000 + 1001 + 1100 + 1101$$

$$0010 - 0011 + 0110 - 0111 + 1010 - 1011 + 1110 - 1111$$

$$0000 + 0001 - 0100 - 0101 - 1000 - 1001 + 1100 + 1101$$

$$0010 - 0011 - 0110 + 0111 - 1010 + 1011 + 1110 - 1111)$$

$$= \frac{1}{2\sqrt{2}} (|0000\rangle + |0001\rangle + |1100\rangle + |1101\rangle + |0010\rangle - |0011\rangle + |1110\rangle - |1111\rangle)$$

|       |       |       |
|-------|-------|-------|
| $q_1$ | $q_3$ | $q_2$ |
|       | $q_4$ |       |
|       |       |       |

$q_3 = q_2$   
 $\therefore$  Consistent

Scenario 3,4  $(1,0), (1,1)$  are the same as  $(0,0), (0,1)$  respectively.

For  $(0,2), (1,2), (2,0), (2,1)$

Since  $\sigma_x \sigma_z, \sigma_z \sigma_x$  eigenvalues =  $\{1, -1\}$

$\therefore$  Can not be measured (Not Hermitians i.e. eigenvalues  $\notin \mathbb{R}$ )

Then how should we proceed?

$$C_3 \downarrow$$

|                              |                              |                              |
|------------------------------|------------------------------|------------------------------|
| $\sigma_z \otimes \sigma_z$  | $\sigma_x \otimes \sigma_z$  | $\sigma_x \otimes \sigma_x$  |
| $\sigma_z \otimes \sigma_x$  | $\sigma_z \otimes \sigma_z$  | $\sigma_z \otimes \sigma_x$  |
| $-\sigma_z \otimes \sigma_z$ | $-\sigma_x \otimes \sigma_x$ | $-\sigma_y \otimes \sigma_y$ |

$r_3 \rightarrow$

In order to satisfy the intersect requirement

Since  $x_{13}, x_{23}, x_{31}, x_{32}$  are calculated from their respective row and column

ex. For  $x_{32}$

|  |          |  |
|--|----------|--|
|  | $x_{12}$ |  |
|  | $x_{22}$ |  |
|  | $x_{32}$ |  |

$\therefore$  Parity of Col = odd

$\therefore x_{32} = 0 ; x_{12} + x_{22} = \text{odd}$   
 $x_{32} = 1 ; x_{12} + x_{22} = \text{even}$

∴ Bob will fill  $x_{32}$  in accordance with  $x_{12}, x_{22}$  parity

∴ Alice who will need to fill row 3 must fill  $x_{32}$  with parity of  $x_{12}, x_{22}$  in mind

Reminder:

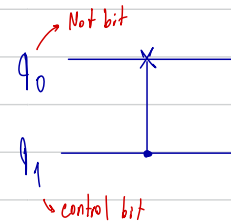
$$\begin{aligned}
 & (H \otimes I) |\Phi^+\rangle = (H \otimes I) \frac{1}{\sqrt{2}} (|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle) \\
 & |\Phi^+\rangle = \frac{1}{\sqrt{2}} [(H \otimes I) (|0\rangle \otimes |0\rangle) + (H \otimes I) (|1\rangle \otimes |1\rangle)] \\
 & = \frac{1}{\sqrt{2}} [(H|0\rangle \otimes I|0\rangle) + (H|1\rangle \otimes I|1\rangle)] \\
 & = \frac{1}{\sqrt{2}} [|+\rangle \otimes |0\rangle + |-\rangle \otimes |1\rangle] \\
 & = \frac{1}{\sqrt{2}} \left[ \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle) + \frac{1}{\sqrt{2}} (|01\rangle - |11\rangle) \right] \\
 & = \frac{1}{2} (|00\rangle + |10\rangle + |01\rangle - |11\rangle) = |\psi_{\text{mixed}}\rangle
 \end{aligned}$$

$$\begin{aligned}
 & (I \otimes H) |\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |01\rangle + |10\rangle - |11\rangle) \\
 & |\Phi^+\rangle = \frac{1}{\sqrt{2}} [(I \otimes H) (|0\rangle \otimes |0\rangle) + (I \otimes H) (|1\rangle \otimes |1\rangle)] \\
 & = \frac{1}{\sqrt{2}} [(I|0\rangle \otimes H|0\rangle) + (I|1\rangle \otimes H|1\rangle)] \\
 & = \frac{1}{\sqrt{2}} [|+\rangle \otimes |+\rangle + |-\rangle \otimes |-\rangle] = \frac{1}{\sqrt{2}} (|++\rangle + |--\rangle) \\
 & = \frac{1}{2} \left[ \frac{1}{\sqrt{2}} (|00\rangle + |01\rangle + |10\rangle + |11\rangle) + \frac{1}{\sqrt{2}} (|00\rangle - |01\rangle - |10\rangle + |11\rangle) \right] \\
 & = \frac{1}{2} \left[ \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle) + \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \right] = |\Phi^+\rangle
 \end{aligned}$$

control  $\rightarrow$  To Flip ( $\sigma_x$ )  
Not to Flip ( $I$ )

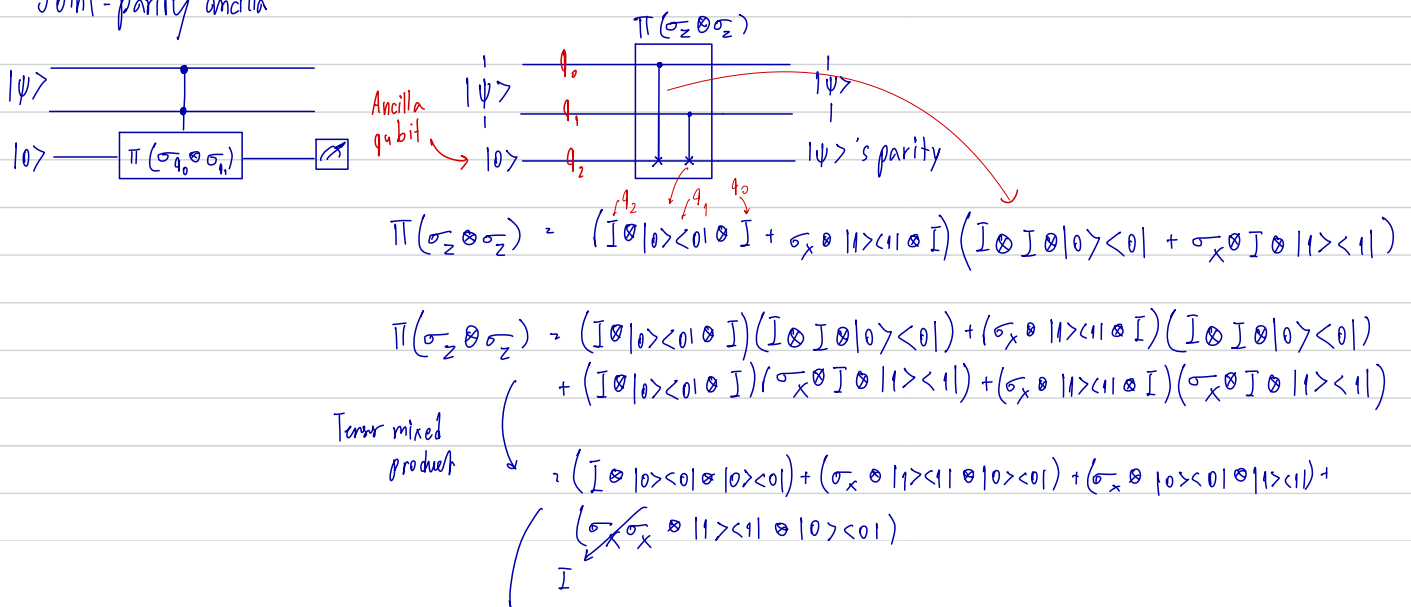
$$\text{CNOT} = \begin{bmatrix} 00 & 01 & 10 & 11 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes \sigma_x$$

ex.  $\rightarrow \text{CNOT}(|01\rangle) = |01\rangle$



$$\begin{aligned}
 & = (|0\rangle\langle 0| \otimes I) (|0\rangle \otimes |1\rangle) + (|1\rangle\langle 1| \otimes \sigma_x) (|0\rangle \otimes |1\rangle) \\
 & \quad \downarrow \text{Mixed product property} \\
 & = (|0\rangle\langle 0| \otimes I) (I|1\rangle) + (|1\rangle\langle 1| \otimes \sigma_x) (\sigma_x|1\rangle) \\
 & = (|0\rangle\langle 0| \otimes I) (|1\rangle) + 0(|1\rangle \otimes |0\rangle) \\
 & = |01\rangle + 0 \\
 & = |01\rangle
 \end{aligned}$$

Joint-parity ancilla



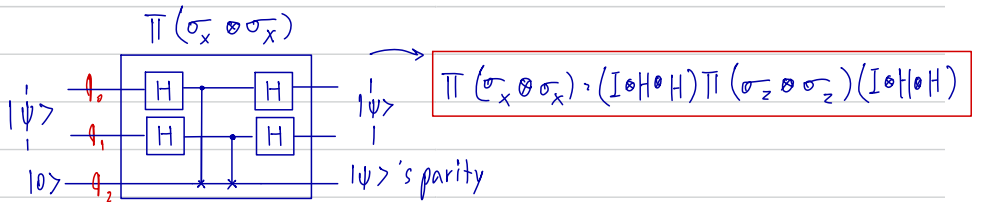
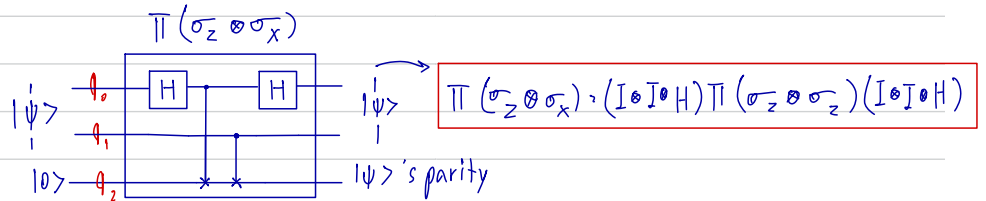
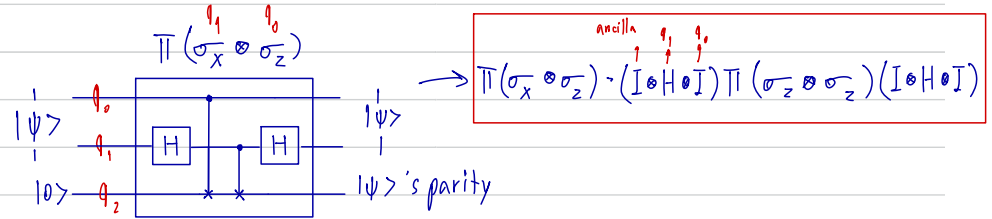
$$\therefore |a\rangle\langle b| \otimes |c\rangle\langle d| \\ |ac\rangle\langle bd|$$

$$\Pi(\sigma_z \otimes \sigma_z) = I \otimes |00\rangle\langle 00| + I \otimes |11\rangle\langle 11| + \sigma_x \otimes |01\rangle\langle 01| + \sigma_x \otimes |10\rangle\langle 10|$$

$q_3 \rightarrow q'_3$   
 If  $00 : |0\rangle \xrightarrow{I} |0\rangle$   $\therefore \Pi(\sigma_z \otimes \sigma_z)$  is the joint-parity function  
 $11 : |0\rangle \xrightarrow{I} |0\rangle$   
 $01 : |0\rangle \xrightarrow{\sigma_x} |1\rangle$   $\Pi(00) = 0$   $\Pi(01) = 1$   
 $10 : |0\rangle \xrightarrow{\sigma_x} |1\rangle$   $\Pi(11) = 0$   $\Pi(10) = 1$

For  $\Pi(\sigma_x \otimes \sigma_z)$ ,  $\Pi(\sigma_z \otimes \sigma_x)$ ,  $\Pi(\sigma_x \otimes \sigma_x)$

Apply change of basis before parity and apply inverse change of basis back so the  $|\psi\rangle$  remain the same i.e.



Row 0, Col 2 (0,2):  $|\psi\rangle = \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle) = |\phi^+\rangle \otimes |\phi^+\rangle$

$$U|_{\text{ancilla}} + |\psi\rangle = (I \otimes I \otimes I \otimes I \otimes I)(|0\rangle \otimes |0\rangle \otimes |\psi\rangle)$$

$(q_1 + q_3) \div 2$   
 $\Pi(\sigma_x, \sigma_z) U|_{\text{ancilla}} + |\psi\rangle = (I \otimes I \otimes I \otimes H \otimes I \otimes I) \Pi(\sigma_z \otimes \sigma_z) (I \otimes I \otimes I \otimes H \otimes I \otimes I) (I \otimes I \otimes I \otimes I \otimes I)(|0\rangle \otimes |0\rangle \otimes |\psi\rangle)$   
 $= (I \otimes I \otimes I \otimes H \otimes I \otimes I) \Pi(\sigma_z \otimes \sigma_z) (I|0\rangle \otimes I|0\rangle \otimes (H \otimes H)|\phi^+\rangle \otimes (I \otimes I)|\phi^+\rangle)$   
 $\therefore (H \otimes H)|\phi^+\rangle \otimes |\phi^+\rangle$   
 $= (I \otimes I \otimes I \otimes H \otimes I \otimes I) \Pi(\sigma_z \otimes \sigma_z) (|0\rangle \otimes |0\rangle \otimes |\phi^+\rangle \otimes |\phi^+\rangle)$   
 $= (I \otimes I \otimes I \otimes H \otimes I \otimes I)$

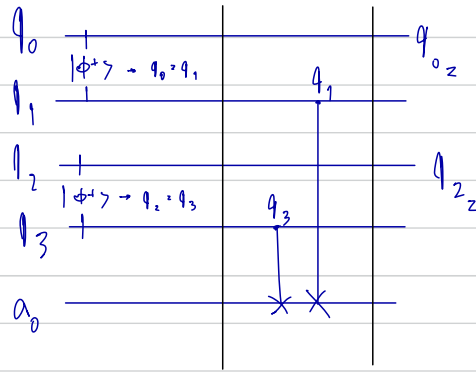
|                             |                             |                             |
|-----------------------------|-----------------------------|-----------------------------|
| $\sigma_z \otimes \sigma_z$ | $\sigma_x \otimes \sigma_z$ | $\sigma_x \otimes \sigma_x$ |
|                             |                             |                             |
|                             |                             |                             |
|                             |                             |                             |

$\text{ancilla}_2 = \Pi(q_{2x}, q_{0z})$   
 $\text{ancilla}_3 = \Pi(q_{2z}, q_{0x})$

Row 2, Col 0 (2,0):

|                              |                              |  |
|------------------------------|------------------------------|--|
| $\sigma_z \otimes \sigma_z$  |                              |  |
| $\sigma_z \otimes \sigma_x$  |                              |  |
| $-\sigma_x \otimes \sigma_z$ | $-\sigma_x \otimes \sigma_x$ |  |

$$(q_3 + q_0 + 1) \cdot 2$$



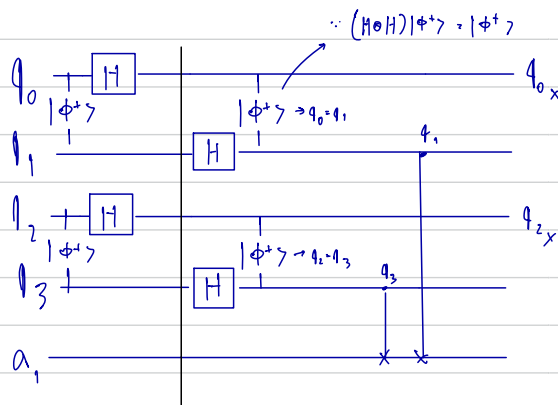
| $(q_0, q_2)$ | $(q_0 + q_2 + 1) \cdot 2$ | $(q_1, q_3)$ | $a_0$ | $(a_0 + 1) \cdot 2$ |
|--------------|---------------------------|--------------|-------|---------------------|
| 0 0          | 1                         | 0 0          | 0     | 1                   |
| 0 1          | 0                         | 0 1          | 1     | 0                   |
| 1 0          | 0                         | 1 0          | 1     | 0                   |
| 1 1          | 1                         | 1 1          | 0     | 1                   |

$$(q_0 + q_1 + 1) \cdot 2 = (a_0 + 1) \cdot 2$$

Row 2, Col 1 (2,1):

|                              |                              |  |
|------------------------------|------------------------------|--|
| $\sigma_x \otimes \sigma_z$  |                              |  |
| $\sigma_z \otimes \sigma_x$  |                              |  |
| $-\sigma_z \otimes \sigma_z$ | $-\sigma_x \otimes \sigma_x$ |  |

$$(q_3 + q_0 + 1) \cdot 2$$

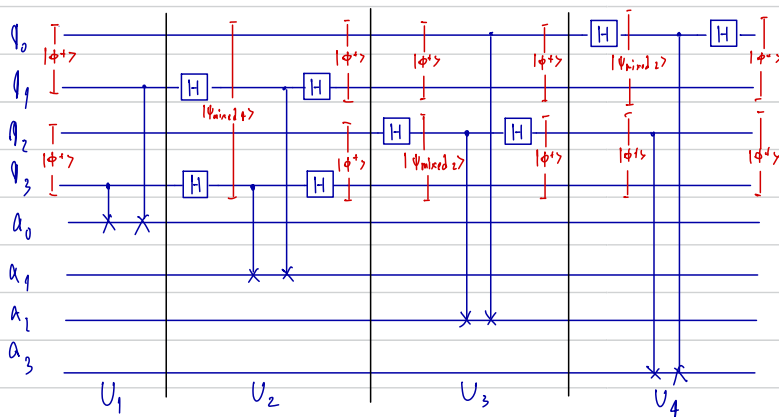


| $(q_0, q_2)$ | $(q_0 + q_2 + 1) \cdot 2$ | $(q_1, q_3)$ | $a_1$ | $(a_1 + 1) \cdot 2$ |
|--------------|---------------------------|--------------|-------|---------------------|
| 0 0          | 1                         | 0 0          | 0     | 1                   |
| 0 1          | 0                         | 0 1          | 1     | 0                   |
| 1 0          | 0                         | 1 0          | 1     | 0                   |
| 1 1          | 1                         | 1 1          | 0     | 1                   |

$$(q_0 + q_2 + 1) \cdot 2 = (a_1 + 1) \cdot 2$$

Row 2, Col 2 (2,2):

|                              |                              |                              |
|------------------------------|------------------------------|------------------------------|
|                              |                              | $\sigma_x \otimes \sigma_z$  |
|                              |                              | $\sigma_z \otimes \sigma_x$  |
| $-\sigma_z \otimes \sigma_z$ | $-\sigma_x \otimes \sigma_x$ | $-\sigma_y \otimes \sigma_y$ |



$$U_1 |a_0\rangle \psi = \left[ (I \otimes |0\rangle\langle 0| \otimes I \otimes |0\rangle\langle 0| \otimes I) + \frac{1}{2} [ |0\rangle\langle 0| \otimes (|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle) ] \right] (H \otimes I \otimes H \otimes I) | \phi^+ \otimes | \phi^+ \rangle \cdot (H \otimes I) | \phi^+ \rangle \otimes (H \otimes I) | \phi^+ \rangle$$

$$= \frac{1}{4} \begin{bmatrix} 0000 + 0001 + 0010 + 0011 \\ + 0100 + 0101 + 0110 + 0111 \\ + 1000 + 1001 + 1010 + 1011 \\ - 1100 - 1101 - 1110 - 1111 \end{bmatrix} = | \psi_{mixed} \rangle$$

$$= \frac{1}{2} ( |0\rangle \otimes |0000\rangle ) + \frac{1}{2} ( |1\rangle \otimes |0011\rangle ) + \frac{1}{2} ( |1\rangle \otimes |1100\rangle ) + \frac{1}{2} ( |0\rangle \otimes |1111\rangle )$$

Intersect table :

- Alice's qubit

- Bob's qubit

main qubit =  $|q_3 q_2 q_1 q_0\rangle$

ancilla qubit =  $|a_3 a_2 a_1 a_0\rangle$

|                                                                      |                                                                      |                                                                        |
|----------------------------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------|
| $(0,0)$<br>$q_1 = q_0$                                               | $(0,1)$<br>$q_3 = q_2$                                               | $(0,2)$<br>$(q_1 + q_3) \times 2$<br>$=$<br>$a_2$                      |
| $(1,0)$<br>$q_3 = q_2$                                               | $(1,1)$<br>$q_1 = q_0$                                               | $(1,2)$<br>$(q_1 + q_3) \times 2$<br>$=$<br>$a_3$                      |
| $(2,0)$<br>$(a_0 + 1) \times 2$<br>$=$<br>$(q_1 + q_0 + 1) \times 2$ | $(2,1)$<br>$(a_1 + 1) \times 2$<br>$=$<br>$(q_2 + q_0 + 1) \times 2$ | $(2,2)$<br>$(q_0 + q_1) \times 2$<br>$=$<br>$(a_2 + a_3 + 1) \times 2$ |